

OCR & AI Extraction Technology Research

MIT ACT — Crennotech Group Project (Smart Document Scan AI) · Week 2 · Individual task (rotating) · 31 July 2026

Why we're looking at this

Before we design the extraction pipeline in Week 4, we need to know what tools exist for reading text out of PDFs, Word docs, Excel files and scanned images — and for pulling out the specific fields we need.

Options we looked at

Tool	What it does	Good for	Watch out for
Tesseract OCR	Free, open-source text recognition from images	Scanned/image PDFs	Needs clean scans; less accurate on messy layouts
Google Document AI	Cloud OCR + structured field extraction	Forms with varying layouts	Paid service; sends data to Google's cloud
AWS Textract	Cloud OCR + table/form extraction	Tables (task lists, frequencies)	Paid service; needs AWS account
Azure Form Recognizer	Cloud OCR + custom form models	Training a model on our own SOW samples	Paid service; needs enough sample documents to train
GPT-based extraction (e.g. via API)	Reads text and pulls out fields using plain instructions	Handling messy or inconsistent wording	Cost per document; needs careful checking for mistakes

Early recommendation

- Use OCR (like Tesseract or a cloud OCR service) first, to turn scanned/image documents into plain text.
- Then use an AI/language model step to read that text and map it onto our standard fields, since customer wording varies so much.
- Add a confidence score to each extracted field, so staff can quickly check anything the system isn't sure about (this will be designed properly in Week 3).

Still to confirm

Final tool choice depends on cost, and on how much control Crennotech wants over where the data is processed. This will be revisited when we design the system architecture in Week 4.